Week 3 Lab — Campus Buddy KICT Starts: The Profile Screen

Time: ~90 min · **You need:** Week 2 done (Flutter installed, emulator works)

Goal: start the app you will build on every week until Week 14. Today: one screen — your profile.

> ***How to use this sheet:*** *do the steps in order, don't skip.* ***CHECK:*** *= what you should see after a step. If it doesn't match, stop and fix before moving on.* ★ *CHECKPOINT = show your demonstrator.*

A. Get the class project (5 min)

Step 1. Open Terminal. Go to the folder where you keep your coursework. Type (Section 2 students: use `section2` in the last line):

```
git clone https://github.com/AhammadAliPK/campus_buddy_kict.git
cd campus_buddy_kict
git checkout section1
flutter pub get
```

This downloads the class project — the app you will build on every week until Week 14. Never used git before? The repo's README (open it on GitHub) explains everything step by step. You will only ever edit `lib/`, `test/`, and `pubspec.yaml`.

Step 2. Start your emulator, then type:

```
flutter run
```

The first run is slow. Wait for it.

CHECK: You see an app with a big `0` and a purple `+` button. Tapping `+` counts up.

★ **CHECKPOINT 1** — the counter app runs on your emulator.

Keep this Terminal open the whole lab. You will press these keys a lot:

Key	What it does
`r`	reload your code (fast, use it always)
`R`	restart the app (use when you change `main.dart`)
`q`	close the app

Step 3. Open `lib/main.dart`. Delete everything in the file. Type this:

```
import 'package:flutter/material.dart';

void main() => runApp(const CampusBuddyKictApp());
```

```
class CampusBuddyKictApp extends StatelessWidget {
  const CampusBuddyKictApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Campus Buddy KICT',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.teal),
      ),
      home: const ProfileScreen(),
    );
  }
}
```

`ProfileScreen` will show red — it doesn't exist yet. That's fine. You create it in Step 4. Don't press anything yet.

TIP: This file is the app's front door. `home:` says which screen opens first.

B. The profile screen (25 min)

Step 4. Make a new file next to `main.dart`: `lib/profile_screen.dart`. Type:

```
import 'package:flutter/material.dart';

class ProfileScreen extends StatefulWidget {
  const ProfileScreen({super.key});

  @override
  State<ProfileScreen> createState() => _ProfileScreenState();
}

class _ProfileScreenState extends State<ProfileScreen> {
  String _nickname = 'Guest Student';
  String _bio =
      'Kuliyyah of ICT · Year 2 · Loves nasi ayam and clean code.';

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('My Profile')),
      body: SingleChildScrollView(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            const CircleAvatar(
              radius: 48,
              backgroundImage: NetworkImage(
                'https://picsum.photos/200',   // placeholder photo service
              ),
            ),
```

```
            const SizedBox(height: 12),
            Text(
              _nickname,
              style: Theme.of(context).textTheme.headlineSmall,
            ),
            const SizedBox(height: 6),
            Text(
              _bio,
              textAlign: TextAlign.center,
              style: const TextStyle(color: Colors.black54),
            ),
          ],
        ),
      ),
    );
  }
}
```

Step 5. In `main.dart`, add this line at the very top, under the first import:

```
import 'package:campus_buddy_kict/profile_screen.dart';
```

Step 6. Press `R` in Terminal (big R — you changed `main.dart`).

CHECK: ★ **CHECKPOINT 2** — a new screen: a bar saying *My Profile* on top, a round photo, your name in big text, grey text under it.

TIP: Simple meanings of what you typed:

- `Scaffold` = a ready-made screen (a top bar + a body).
- `Column` = put things top-to-bottom.
- `SingleChildScrollView` = let the screen scroll. It matters when the keyboard opens (Part C).
- `_nickname` and `_bio` = the screen's **data**. Nothing can change them *yet* — that's Part C.

TIP: The photo comes from the internet. No internet = empty grey circle. If that happens in the lab: instead of `NetworkImage(...)`, use a photo of yourself — make a folder `assets/`, put `me.jpg` in it, add under `flutter:` in `pubspec.yaml`:

```
  assets:
    - assets/me.jpg
```

and write `AssetImage('assets/me.jpg')` instead. (After editing pubspec.yaml, stop the app with `q` and `flutter run` again — reload is not enough.)

Try it yourself (5 min)

a. In the bio `Text`, add `maxLines: 1, overflow: TextOverflow.ellipsis`. Make the bio very long. Press `r`. What happened to the extra words? b. Put `Center(` around the `Column`. Press `r`. Why is scrolling still better here? (Undo the `Center` after.)

C. Make it editable (35 min)

Step 7. In `profile_screen.dart`, delete **everything** in the file and paste this complete new version:

```
import 'package:flutter/material.dart';

class ProfileScreen extends StatefulWidget {
  const ProfileScreen({super.key});

  @Override
  State<ProfileScreen> createState() => _ProfileScreenState();
}

class _ProfileScreenState extends State<ProfileScreen> {
  String _nickname = 'Guest Student';
  String _bio =
      'Kuliyyah of ICT · Year 2 · Loves nasi ayam and clean code.';

  final _nicknameCtrl = TextEditingController();
  final _bioCtrl = TextEditingController();
  bool _editing = false;

  @override
  void initState() {
    super.initState();
    _nicknameCtrl.text = _nickname;   // pre-fill from current state
    _bioCtrl.text = _bio;
  }

  @override
  void dispose() {
    _nicknameCtrl.dispose();
    _bioCtrl.dispose();
    super.dispose();
  }

  void _toggleEdit() {
    setState(() {
      if (_editing) {
        // leaving edit mode: copy controller values into state
        _nickname = _nicknameCtrl.text.trim();
        _bio = _bioCtrl.text.trim();
      }
      _editing = !_editing;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('My Profile'),
        actions: [
          IconButton(
```

```
          icon: Icon(_editing ? Icons.check : Icons.edit),
          onPressed: _toggleEdit,
        ),
      ],
    ),
    body: SingleChildScrollView(
      padding: const EdgeInsets.all(16),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.stretch,
        children: [
          const CircleAvatar(
            radius: 48,
            backgroundImage: NetworkImage('https://picsum.photos/200'),
          ),
          const SizedBox(height: 12),
          if (_editing) ...[
            TextField(
              controller: _nicknameCtrl,
              decoration: const InputDecoration(
                labelText: 'Nickname',
                border: OutlineInputBorder(),
                prefixIcon: Icon(Icons.person),
              ),
            ),
            const SizedBox(height: 12),
            TextField(
              controller: _bioCtrl,
              maxLines: 3,
              keyboardType: TextInputType.multiline,
              decoration: const InputDecoration(
                labelText: 'Bio',
                border: OutlineInputBorder(),
              ),
            ),
            const SizedBox(height: 12),
            ElevatedButton.icon(
              onPressed: _toggleEdit,
              icon: const Icon(Icons.save),
              label: const Text('Save'),
            ),
          ] else ...[
            Text(_nickname,
                style: Theme.of(context).textTheme.headlineSmall,
                textAlign: TextAlign.center),
            const SizedBox(height: 6),
            Text(_bio,
                textAlign: TextAlign.center,
                style: const TextStyle(color: Colors.black54)),
          ],
        ],
      ),
    ),
  );
```

```
  }
}
```

Step 8. Press `r` in Terminal.

Step 9. Try it: tap the ═■ in the top bar. The view becomes a form with your current text inside. Change the text. Tap OK.

CHECK: ★ **CHECKPOINT 3** — the new text stays on screen after saving. Tap ═■ again — the form starts with the new text.

TIP: This loop is the heart of every app you will ever make:

```
you type  ->  controller  ->  Save copies into state  ->  screen redraws
```

TIP: Small but important things in the code:

- A `TextEditingController` is the box a `TextField` types into. `_nicknameCtrl.text` is whatever is typed right now.
- `setState(...)` = "data changed, redraw the screen". Without it the screen would not update. (Week 2 rule.)
- `if (_editing) ...[ ] else ...[ ]` = show the form OR the view. The three dots `...` insert many widgets at once.
- `initState` runs once when the screen opens (fill the boxes). `dispose` runs when the screen dies (throw away the boxes).

D. Exercise in class (10 min)

Add a **Reset** `TextButton` under the Save button. Pressing it puts everything back: nickname = `'Guest Student'`, bio = the original text. (Remember to fix the boxes too: `_nicknameCtrl.text = ...`, so the next edit starts clean.)

E. Save your work with git (5 min)

You **cloned** this project — git is already set up for you. No `git init`, nothing to create. You just save your work with a commit.

Step 10. In Terminal press `q` (close the app). Then:

```
git add .
git commit -m "week 3 profile screen"
```

CHECK: Terminal prints a commit line with your files.

TIP: This commit lives on **your laptop only**. You cannot push to the lecturer's branches (`main`, `section1`, `section2` — GitHub blocks you). Each week's class code arrives with `git pull`. If `git pull` ever shows an error mentioning "conflict" — stop and ask, never guess.

Step 11. Open `campus_buddy_kict/.gitignore`. It came with the class repo. It tells git to skip junk files like `build/` and `.dart_tool/` (they are rebuilt every run — no need to save them).

You change nothing this week. Just remember:

- git saves: `lib/`, `test/`, `pubspec.yaml`, `assets/`
- git skips: `build/`, `.dart_tool/`, machine files
- If a later week needs a new line in `.gitignore`, that lab will show you the exact line.

Homework (Assignment 2, graded)

1. Add a **matric number**: make `String _matric = '202300000';` in the state, show it as small grey text under the bio, and add a box for it in the form with `keyboardType: TextInputType.number`.
2. If someone saves with an **empty** nickname, keep `'Guest Student'` instead. (Hint: `''.trim().isEmpty` is true for empty or spaces-only.)
3. Put your own photo in `assets/` (like Step 4's tip) and use it instead of the internet photo. Keep it — this photo is your app's face for the rest of the semester.

If something goes wrong

Problem	Fix
"used after disposed" error	`dispose()` must be inside `_ProfileScreenState`, not `ProfileScreen`
Saved text has extra spaces	Use `.trim()` when reading the controller
Keyboard covers the Save button	You deleted `SingleChildScrollView` — put it back
pubspec.yaml error	Spaces matter — use exactly 2 per level. Then `q` and `flutter run` again
Nothing changes after `r`	You edited `main.dart` — press `R` instead
Flutter opened a desktop window, not the emulator	`flutter devices` shows the list, then `flutter run -d <id>`
Red screen: `ProfileScreen isn't defined`	The import from Step 5 is missing or has a typo

Next lab: a *My Courses* list — all your subjects on one screen. Bring your real timetable (at least 5 courses).